Note in Proximal algorithm for nonsmooth function involving analytic features

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Abstract

This note mainly refer to [1]. This paper mainly describes how to prove pointwise convergence on x in proximal algorithm under Lojasiewicz condition but without Lipschitz smooth.

The proof techniques mentioned in the article are worth learning include $-s^{1-\theta}$ is convex. Consider different situations then conclude a conclusion, i.e., we can divide the problem into two situations $|x^{k+1} - x^k| \geq r|x^k - x^{k-1}|$ and $|x^{k+1} - x^k| < r|x^k - x^{k-1}|$ in page 12 and $h(\Delta_k) \leq Rh(\Delta_{k-1})$,$h(\Delta_k) > Rh(\Delta_{k-1})$ in page 14.

Keywords: Lojasiewicz condition, non-smooth function

1 Preliminaries

1.1 Proximal algorithm

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper lower semicontinuous function. Given $x^0 \in \mathbb{R}^n$, we consider the following discrete dynamical system

$$x^{k+1} \in argmin \left\{ f(u) + \frac{1}{2\lambda_k} \left\| u - x^k \right\|^2 \right\} \tag{1}$$

where λ_k is a positive sequence.

Necessary and sufficient conditions for this algorithm to be well-defined can be found in Rockafellar and Wets[Variational analysis exercise 1.24]. We simply assume here that.

1. $(\mathcal{H}_1)$ $\inf_{\mathbb{R}^n} f > -\infty$
2. $(\mathcal{H}_2)$The restrication of f to its domain is a continuous function(on dom f).

Let us fix some positive parameters λ_- and λ_+ with $0 < \lambda_- < \lambda_+ < \infty$. From now on we assume that $\lambda_k \in (\lambda_-, \lambda_+)$ for all $k \in \mathbb{N}$

2 Basic Property

Proposition 2.1. *Let (x^k) be a sequence which generated by proximal algorithm, and denote by $\omega(x_0)$ the set of its limit points. Then*

1. *The sequence $(f(x^k))$ is nonincreasing.*
2. $\sum |x^{k+1} - x^k|^2 < +\infty$
3. *If f satisfies $(\mathcal{H}_2)$ then $\omega(x^0) \subset crit f$*
4. *If f satisfies $(\mathcal{H}_2)$ and the sequence x^k is bounded then $\omega(x^0)$ is a nonempty compact connected set, and*
$$d(x^k, \omega(x^0) \to 0 \qquad as\ k \to \infty$$
5. *If f satisfies $(\mathcal{H}_2)$ then f is finite and constant on $\omega(x^0)$.*

Sketch of the proof. By definition, we have
$$f(x^{k+1}) + \frac{1}{2\lambda_k}[x^{k+1} - x^k]^2 \leq f(x^k) \tag{2}$$
This means that $f(x^k)$ is nonincreasing(1) and by summing the inequalities we have
$$\sum_{k=0}^{N} |x^{k+1} - x^k|^2 \leq 2\lambda_+[f(x^0) - f(x^{N+1})] \leq 2\lambda_+[f(x^0) - \inf f] < \infty \tag{3}$$
Then we complete the proof of (2).

Let us deal with 5 and 3 in proposition 2.1. For any limit point $\bar{x}$ of f. we can use the lower semicontinutity of f to obtain that $\lim_{k\to\infty} \geq f(\bar{x})$. If, in addition, f satisfies $(\mathcal{H}_2)$ then the above inequality is actually an equality and (5) is proved. By using 2 , $x^{k+1} = x^k - \lambda_k g^{k+1}$, and the fact that $\lambda_k \geq \lambda_- > 0$ we can assume that there exists $k_p \to \infty$ such that $\left\{(x^{k_p}, g^{k_p})\right\} \to (\bar{x}, 0)$ with $f(x^{k_p}) \to f(\bar{x})$. Owing to the definition of the limiting subdifferential it follows that $(\bar{x}, 0)$ belongs to the graph of ∂f, so 3 is proved.

Item 4 follows 2 together with some classical properties of sequences in $\mathbb{R}^n$. □

Remark: Even when x^k is bounded, the Convergence of the whole sequence x^k may fail even for a finite-valued smooth function f, see Palis and De Melo([6]) or Absil([5]).

3 Convergence analysis

Definition 3.1 (Lojasiewicz inequalities)**.** Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper lower semicontinuous function that satisfies $(\mathcal{H}_2)$. The function is said to have the Lojasiewicz property if: For any limiting-critical point $\hat{x}$, that is $\hat{x} \in \text{crit} f$, there exist C, $\epsilon > 0$ and $\theta \in [0, 1)$ such that
$$|f(x) - f(\hat{x})|^\theta \leq C|x^*| \quad \forall x \in B(\hat{x}, \epsilon), \quad \forall x^* \in \partial f(x) \tag{4}$$

Remark. Kurdyka-Lojasiewicz condition is a special case with $\theta = \frac{1}{2}$.

Lemma 3.1. *Assume that f has the Lojasiewicz preperty*

1. *If K is a connected subset of the set of critical points of f, then f is constant on K.*
2. *If in addition K is a compact set, then there exist C, $\epsilon > 0$ and $\theta \in [0, 1)$ such that*

$$\forall x \in \mathbb{R}^n,\ d(x, K) \leq \epsilon,\ \forall x^* \in \partial f(x),\ |f(x) - f(\hat{x})|^\theta \leq C|x^*| \tag{5}$$

where $\hat{x}$ is any limiting-critical point, $\hat{x} \in crit f$

Proof. Item 1 is a straightforward consequence of 2, let us therefore deal with 2. The compact set K can be covered by **a finite number** of open balls $B(x_i, \epsilon_i)$, with $x_i \in K(i = 1, \ldots, p)$ on which 4 holds with constants C_i, θ_i. In other words, for each $i \in \{1, \ldots, p\}$ and for each $x \in B(x_i, \epsilon_i)$ we have

$$|f(x) - f(x_i)|^{\theta_i} \leq C_i|x^*| \tag{6}$$

for all $x^* \in \partial f(x)$. As a consequence, f is locally constant (and continuous) on the connected set K, it is therefore constant. By choosing $\epsilon > 0$ sufficiently small, we obtain that

$$\{x \in \mathbb{R}^n : d(x, K) \leq \epsilon\} \subset \cup_{i=1}^p B(x_i, \epsilon_i) \tag{7}$$

and the claimed result holds by letting $C = maxC_i$ and $\theta = max\theta_i$.

Example 3.1. Real-analytic functions have the Lojasiewicz property, see [[4]]

Example 3.2. An interesting class of functions satisfying the Lojasiewicz property is given by **semialgerbraic** functions. These are functions whose graphs can be expressed as

$$\bigcup_{i=1}^{p} \bigcap_{j=1}^{q} \left\{x \in \mathbb{R}^n : P_{ij}(x) = 0, Q_{ij} > 0\right\} \tag{8}$$

where for all $1 \leq i \leq p, 1 \leq j \leq q$ the $P_{ij}Q_{ij} : \mathbb{R}^n \mapsto \mathbb{R}$ are polynomial functions.

Example 3.3. Convex functions satisfying the following growth conditions

$$\begin{gathered} \forall \hat{x} \in argmin\, f\ \exists C > 0, r \geq 1, \epsilon > 0, \forall x \in B(\hat{x}, \epsilon) \\ f(x) \geq f(\hat{x}) + C\, d(x, argmin\, f)^r \end{gathered} \tag{9}$$

Comply with 4 (with $\theta = 1 - \frac{1}{r}$),see [bolte2007lojasiewicz] . This is in particular the case of strongly convex functions, that is proper lower semicontinuosu functions $f : \mathbb{R}^n \to \mathbb{R}\{+\infty\}$ which satisfy

$$\langle x^* - y^*, x - y\rangle \geq k|x - y|^2 \forall (x, x^*), (y, y^*) \in \text{graph}\, \partial f \tag{10}$$

Example 3.4. Infinite-dimensional versions of 4 have been developed in view of the asymptotic analysis of dissipative evolution equations. These can be found in Simon[[7]], and Haraux[[2]].

Example 3.5. o-minimal structure[[3]]

□

4 Convergence analysis

Theorem 4.1. *Assume that f satisfies ($\mathcal{H}_1$),($\mathcal{H}_2$),($\mathcal{H}_3$) and let $(x^k)_{k\in\mathbb{N}}$ be a sequence generated by the proximal algorithm. If the sequence is bounded then we have*

$$\sum_{k=0}^{+\infty} |x^{k+1} - x^k| < \infty \tag{11}$$

in particular the whole sequence $(x^k)_{k\in\mathbb{N}}$ converges to some critical point of f.

Proof. Changing f into $f - inf_{k\geq 0} f(x^k)$ we can assume with no loss of generality that $f(x^k)$ converges to 0. The case when $x^{k+1} = x^k$ for some $k \geq 1$ has no consequence on the asymptotic analysis, so that we may suppose that $|x^{k+1} - x^k| > 0$ for all $k \geq 0$. In view of 4, we obtain also that $f(x^k)$ is positive and decreases(strictly) to 0.

By using the convex inequality for the function $s > 0 \to -s^{1-\theta}$ and 4 for all $k \geq 0$ we obtain that

$$\begin{aligned} f(x^k)^{1-\theta} - f(x^{k+1})^{1-\theta} &\geq (1-\theta) f(x^k)^{-\theta}(f(x^k) - f(x^{k+1})) \\ &\geq (1-\theta) f(x^k)^{-\theta} \frac{1}{2\lambda_k} |x^{k+1} - x^k|^2 \end{aligned} \tag{12}$$

By proposition 2.1 (3) and (4), and 3.1 (take $K = \omega(x_0)$) there exist an integer N_0, real numbers C and $\theta \in (0,1)$ such that

$$0 < |f(x^k)|^\theta \leq C|g^k| = \frac{C}{\lambda_k}|x^k - x^{k-1}| \ \forall k \geq N_0 \tag{13}$$

Combining the above result with 12 (recall $\lambda_k > \lambda_- > 0$)yields the existence of a positive constant $M > \frac{C}{1-\theta}$ (Combine with 12 and 13) such that

$$\frac{|x^{k+1} - x^k|^2}{|x^k - x^{k-1}|} \leq M(f(x^k)^{1-\theta} - f(x^{k+1})^{1-\theta}) \tag{14}$$

Fix $r \in (0,1)$ and take $k \geq N_0$. There are two case

1. If $|x^{k+1} - x^k| \geq r|x^k - x^{k-1}|$,then 14 implies that

$$|x^{k+1} - x^k| \leq \frac{M}{r}[f(x^k)^{1-\theta} - f(x^{k+1})^{1-\theta}] \tag{15}$$

2. If $|x^{k+1} - x^k| < r|x^k - x^{k-1}|$, obvious that we have $|x^{k+1} - x^k| < r|x^k - x^{k-1}|$.

Combine both of them, we have

$$|x^{k+1} - x^k| \leq r|x^k - x^{k-1}| + \frac{M}{r}[f(x^k)^{1-\theta} - f(x^{k+1})^{1-\theta}] \tag{16}$$

Then we have an induction

$$\begin{aligned} |x^{N_0+1} - x^{N_0}| &\leq r|x^{N_0} - x^{N_0-1}| + \frac{M}{r}(f(x^{N_0})^{1-\theta} - f(x^{N_0+1})^{1-\theta}) \\ |x^{N_0+2} - x^{N_0+1}| &\leq r|x^{N_0+1} - x^{N_0}| + \frac{M}{r}(f(x^{N_0+1})^{1-\theta} - f(x^{N_0+2})^{1-\theta}) \\ &\leq r^2|x^{N_0} - x^{N_0-1}| + r \cdot \frac{M}{r}(f(x^{N_0})^{1-\theta} - f(x^{N_0+1})^{1-\theta}) + \frac{M}{r}(f(x^{N_0+1})^{1-\theta} - f(x^{N_0+2})^{1-\theta}) \end{aligned} \tag{17}$$

summing up then we have

$$
\begin{aligned}
|x^{N_0+2}-x^{N_0+1}|+|x^{N_0+1}-x^{N_0}| \leq & r|x^{N_0}-x^{N_0-1}|+\frac{M}{r}(f(x^{N_0})^{1-\theta}-f(x^{N_0+1})^{1-\theta}) \\
& +r^2|x^{N_0}-x^{N_0-1}|+r\cdot\frac{M}{r}(f(x^{N_0})^{1-\theta}-f(x^{N_0+1})^{1-\theta})+\frac{M}{r}(f(x^{N_0+1})^{1-\theta}-f(x^{N_0+2})^{1-\theta}) \\
& =r|x^{N_0}-x^{N_0-1}|+r^2|x^{N_0}-x^{N_0-1}| \\
& +r\cdot\frac{M}{r}(f(x^{N_0})^{1-\theta}-f(x^{N_0+1})^{1-\theta})+\frac{M}{r}(f(x^{N_0})^{1-\theta}-f(x^{N_0+2})^{1-\theta})
\end{aligned}
\tag{18}
$$

Consider next iteration, we have

$$
\begin{aligned}
|x^{N_0+3}-x^{N_0+2}| & \leq r|x^{N_0+2}-x^{N_0+1}|+\frac{M}{r}(f(x^{N_0+2})^{1-\theta}-f(x^{N_0+3})^{1-\theta}) \\
& \leq r^3|x^{N_0}-x^{N_0-1}|+r^2\cdot\frac{M}{r}(f(x^{N_0})^{1-\theta}-f(x^{N_0+1})^{1-\theta}) \\
& +r\cdot\frac{M}{r}(f(x^{N_0+1})^{1-\theta}-f(x^{N_0+2})^{1-\theta})+\frac{M}{r}(f(x^{N_0+2})^{1-\theta}-f(x^{N_0+3})^{1-\theta})
\end{aligned}
\tag{19}
$$

we know $f(x^k)^{1-\theta}$ is a monotonically decreasing sequence, i.e. $f(x^{N_0+i})^{1-\theta}-f(x^{N_0+i+1})^{1-\theta}\geq 0$. Therefore, we have

$$
f(x^{N_0+i})^{1-\theta}-f(x^{N_0+i+1})^{1-\theta}\leq f(x^{N_0})^{1-\theta}-f(x^{N+1})^{1-\theta}\ \forall i\in[0,N-N_0] \tag{20}
$$

then we have the conclusion

$$
\sum_{k=N_0}^{N}|x^{k+1}-x^k|\leq\frac{r}{1-r}|x^{N_0}-x^{N_0-1}|+\frac{M}{r(1-r)}[f(x^{N_0})^{1-\theta}-f(x^{N+1})^{1-\theta}] \tag{21}
$$

and the conclusion follows from the fact that f is bounded below. □

Remark. 1. The trick proved here is mainly by $-s^{1-\theta}$ is a convex function

2. It would be interesting to compare the results with those obtained in Combettes and Pennanen under a cohypomonotonicity assumption.[.....].
3. If $(x^k)_{k\in\mathbb{N}}$ is a bounded sequence generated by proximal algorithm, let us denote by x^∞ its(unique) limit point. Applying Proposition 2.1 (3) and 4 we obtain the existance of a neighborhood around x^∞ such that 4 holds. The number θ appearing in 4 is then called a Lojasiewicz exponent of x^∞.

Theorem 4.2 (Rate of convergence)**.** *Assume that f satisfies $(\mathcal{H}_1),(\mathcal{H}_2),(\mathcal{H}_3)$ and let $(x^k)_{k\in\mathbb{N}}$ be a sequence generated by the proximal algorithm. Assume that $(x^k)_{k\in N}$ is bounded and denote by θ a Lojasiewicz exponent of x^∞. The following estimations hold*

1. *If $\theta=0$, the sequence $(x^k)_{k\in\mathbb{N}}$ converges in a finite number of steps.*
2. *If $\theta\in(0,\frac{1}{2}]$ then there exists $c>0$ and $Q[0,1]$ such that*
$$|x^k-x^\infty|\leq cQ^k \tag{22}$$
3. *If $\theta\in(\frac{1}{2},1]$ then there exists $c>0$ such that*
$$|x^k-x^\infty|\leq ck^{-\frac{1-\theta}{2\theta-1}} \tag{23}$$

Proof. Denote $\sum_{p=k}^{\infty}|x^{p+1}-x^p|$ by Δ_k which is finite by Theorem 1. Recall theorem 1, we have

$$\sum_{k=N_0}^{N}|x^{k+1}-x^k| \leq \frac{r}{1-r}|x^{N_0}-x^{N_0-1}| + \frac{M}{r(1-r)}[f(x^{N_0})^{1-\theta}-f(x^{N+1})^{1-\theta}] \tag{24}$$

Since $r \in (0,1)$, $\Delta_{k-1}-\Delta_k = |x^k - x^{k-1}|$, and by the definition of f, $f(x^k) > 0$ for any k. Then we have

$$\begin{aligned}\Delta_k \leq & \frac{1}{1-r}(\Delta_{k-1}-\Delta_k) + \frac{M}{r(1-r)}f(x^k)^{1-\theta} \\ \leq & \frac{1}{1-r}(\Delta_{k-1}-\Delta_k) + \frac{M}{r(1-r)}(C|g^k|)^{\frac{1-\theta}{\theta}} \\ \leq & \frac{1}{1-r}(\Delta_{k-1}-\Delta_k) + (\lambda_-)^{1-\frac{1}{\theta}}\frac{MC^{\frac{1-\theta}{\theta}}}{r(1-r)}(\Delta_{k-1}-\Delta_k)^{\frac{1-\theta}{\theta}}\end{aligned} \tag{25}$$

where the second and third inequality follow respectively from Lojasiewicz preperty and proximal algorithm.

Assume that $\theta \in (\frac{1}{2},1)$, so that $\frac{1-\theta}{\theta} < 1$. Since $\Delta_k \to 0$ as $k \to \infty$, we deduce from 25 that there exist an integer $N_1 \geq N_0$ and a positive constant C_1 such that

$$\Delta_k^{\frac{\theta}{1-\theta}} \leq C_1(\Delta_{k-1}-\Delta_k) \quad \forall k \geq N_1 \tag{26}$$

We omit the first term in 25, since $(\Delta_{k-1}-\Delta_k)$ **is higher order infinitesimal here.**

Let $R \in (1,+\infty)$, take $k > N_1$,$h(s) = s^{-\frac{\theta}{1-\theta}}$ (which is a monotonically decreasing function) and **if** $h(\Delta_k) \leq Rh(\Delta_{k-1})$($\Delta_{k-1} \geq \Delta_k$ then we have $h(\Delta_k) \geq h(\Delta_{k-1})$, so the existence of R is reasonable.)

By rewriting 26 as

$$1 \leq \frac{C_1(\Delta_{k-1}-\Delta_k)}{\Delta_k^{\frac{\theta}{1-\theta}}} \tag{27}$$

we obtain that

$$\begin{aligned}1 \leq & C_1(\Delta_{k-1}-\Delta_k)h(\Delta_k) \\ \leq & RC_1(\Delta_{k-1}-\Delta_k)h(\Delta_{k-1}) \\ \leq & RC_1\int_{\Delta_k}^{\Delta_{k-1}}h(s)ds \\ \leq & RC_1\frac{1-\theta}{1-2\theta}[\Delta_{k-1}^{\frac{1-2\theta}{1-\theta}}-\Delta_k^{\frac{1-2\theta}{1-\theta}}]\end{aligned} \tag{28}$$

Thus if we set $\mu = \frac{2\theta-1}{(1-\theta)RC_1} > 0$ and $v = \frac{1-2\theta}{1-\theta} < 0$ one obtains that

$$0 < \mu \leq \Delta_k^v - \Delta_{k-1}^v \tag{29}$$

If $h(\Delta_k) > Rh(\Delta_{k-1})$(Since $h(\Delta_k) \geq h(\Delta_{k-1})$, WLOG, let R is close to 1.) It follows immediately that $\Delta_k \leq q\Delta_{k-1}$ where $q = (\frac{1}{R})^{\frac{1-\theta}{\theta}} \in (0,1)$ and futhermore, recalling that v is negative we have

$$\begin{aligned}\Delta_k^v \geq & q^v\Delta_{k-1}^v \\ \Delta_k^v - \Delta_{k-1}^v \geq & (q^v-1)\Delta_{k-1}^v\end{aligned} \tag{30}$$

Since $q^v - 1 = (\frac{1}{R})^{\frac{1-\theta}{\theta}\frac{1-2\theta}{1-\theta}} - 1 = (\frac{1}{R})^{\frac{1-2\theta}{\theta}} - 1 > 0$ and $\Delta_p \to 0^+$ as $p \to +\infty$, there exists $\bar{\mu} > 0$ such that $(q^v-1)\Delta_{p-1}^v > \bar{\mu}$ for all $p \geq N_1$. Therefore, we obtain that

$$\Delta_k^v - \Delta_{k-1}^v \geq \hat{\mu} > 0 \quad \forall k \geq N_1. \tag{31}$$

Combine the two cases, we have inequality

$$\Delta_k^v - \Delta_{k-1}^v \geq \hat{\mu} > 0 \quad \forall k \geq N_1. \tag{32}$$

where $\hat{\mu} = min\{\mu, \bar{\mu}\}$ By summing those inequalities from N_1 to some N greater than N_1 we obtain that $\Delta_N^v - \Delta_{N_1}^v \geq \hat{\mu}(N - N_1)$ and consequently 3 follows from

$$\Delta_N \leq \left[\Delta_{N_1}^v + \hat{\mu}(N - N_1)\right]^{\frac{1}{v}} \leq cN^{-\frac{1-\theta}{2\theta-1}} \tag{33}$$

When $\theta \in (0, \frac{1}{2}]$, then $\frac{1-\theta}{\theta} > 1$, 25 shows that Δ_k complies with the following inequality(for k sufficiently large) where C_2 is a positive constant.

$$\Delta_k \leq C_2(\Delta_{k-1} - \Delta_k)$$

We omit the second term in 25, since $(\Delta_{k-1} - \Delta_k)$ **is higher order infinitesimal here.**

Which implies that $\Delta_k \leq \frac{C_2}{1+C_2}\Delta_{k-1}$ and item 2 follows easily with $Q = \frac{C_2}{1+C_2}$ by induction.

Assume that $\theta = 0$, set $I := \{k \in \mathbb{N} : x_{k+1} \neq x_k\}$ and take k in I. If k is sufficiently large we have

$$\frac{1}{\lambda_k^2}|x^{k+1} - x^k|^2 = |g^{k+1}|^2 \geq C_3 > 0 \tag{34}$$

so that 2 implies that

$$f(x^{k+1}) \leq f(x^k) - \frac{1}{2\lambda_k}|x^{k+1} - x^k|^2 \leq f(x^k) - C_3\frac{\lambda_-}{2} \tag{35}$$

Since $f(x_k)$ is known to converge to zero the above inequality clearly implies that I is finite and claim 1 follows immediately. □

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